

DUAL-MOTOR PMSM DYNAMOMETER PLATFORM

CONTROL SOFTWARE & VALIDATION

Embedded control, supervisory behavior, communications, acquisition obligations, verification, and public release boundaries.

ENGINEERING REQUIREMENTS DOCUMENT | REVISION 2.6 | 10 AUGUST 2026

Prepared by	Herder Elektronische Systemen
Document scope	Embedded control, supervisory behavior, communications, acquisition obligations, verification, and public release boundaries.
Evidence basis	Controlled internal bench evidence summarized at black-box level; no source artifacts or reproducible implementation details.
Distribution	Public customer-facing release; implementation IP, source artifacts, setup procedures, and reproducible design detail excluded.
DOCUMENT PURPOSE Defines the public functional, interface, safety, verification, and acceptance requirements for the control and supervisory subsystem without releasing implementation intellectual property.	

Functional Requirements Document Framework

WHY THIS IS AN FRD

This document defines what the subsystem shall do, the externally observable interfaces and constraints under which it shall operate, and the evidence used to verify those obligations. It intentionally omits how the requirements are implemented.

Purpose, Scope, and System Boundary

Embedded control, supervisory behavior, communications, acquisition obligations, verification, and public release boundaries.

Boundary element	Included in this FRD	Explicitly excluded
Control behavior	Drive/load coordination, enable behavior, feedback, communication, acquisition obligations	Control models, code, algorithms, tuning values, callbacks, generated artifacts
Host behavior	Supervisory command, status, and evidence obligations	GUI source, host models, logging scripts, setup/deployment instructions
Characterization	What evidence must be supportable and how claims are bounded	Raw data, equations beyond common engineering definitions, filters, binning, calibration, analysis code

Users and Operational Scenarios

Actor	Primary interaction	Expected outcome
Operator	Command, enable, monitor, stop, and review system status	Controlled operation with observable state
Test engineer	Request approved characterization and review summary evidence	Traceable high-level results without access to implementation IP
Customer / reviewer	Assess requirements, verification status, and limitations	Understand demonstrated capability and release boundary

Functional Requirements

ID	The system shall...	Rationale / explanation	Verification
CTRL-FR-001	coordinate two coupled motor channels so one can establish shaft motion while the other provides controlled loading or assistance.	Defines the core dynamometer function without prescribing implementation.	Bench demonstration
CTRL-FR-002	accept bounded supervisory commands for speed, loading, enable state, and operating selection.	Allows controlled operation from a supervisory interface.	Functional demonstration
CTRL-FR-003	maintain independent feedback for both machines during closed-loop operation.	Independent feedback is necessary for drive/load observability.	Functional demonstration
CTRL-FR-004	provide operator-visible status sufficient to determine whether commanded operation is active and whether feedback is being received.	Supports safe supervision and diagnosability.	Demonstration
CTRL-FR-005	support a supervisory communications path for command and diagnostic exchange.	Enables integration with an external real-time or engineering host.	Interface demonstration
CTRL-FR-006	support acquisition of the quantities required for engineering characterization while retaining implementation details as controlled information.	Preserves evidence capability without releasing analysis IP.	Evidence review
CTRL-FR-007	transition to a bounded non-torque-producing or inhibited state when commanded operation is	Defines a minimum black-box safety behavior.	Functional test

ID	The system shall...	Rationale / explanation	Verification
	disabled.		

Performance and Quality Requirements

ID	Requirement	Acceptance basis
CTRL-NFR-001	Time-critical control shall execute independently of graphical-host refresh timing.	Observed motor regulation does not depend on host display cadence.
CTRL-NFR-002	Public evidence shall be traceable to a controlled internal test record without publishing raw data or scripts.	Published claims can be substantiated privately.
CTRL-NFR-003	The public package shall contain no source code, editable control models, setup instructions, calibration constants, detailed protocol definitions, or reproducible implementation mappings.	Package inspection confirms only black-box documents are present.
CTRL-NFR-004	Claims shall remain limited to bench-demonstrated capability and shall not imply production qualification.	Document review confirms controlled wording.

External Interface Requirements

Interface	Required behavior	Failure / boundary condition
Supervisory command interface	Accepts bounded drive/load/enable requests at a functional level.	Loss or invalid command shall not create an unbounded command state.
Machine feedback interface	Provides independent feedback from both machine channels to the embedded control function.	Missing or invalid feedback is treated as a fault or invalid operating condition.
Diagnostic interface	Exposes sufficient system state for operation and fault review.	Public release omits message formats, identifiers, payload layouts, and signal names.

Interface	Required behavior	Failure / boundary condition
Evidence interface	Produces reviewable characterization outputs under controlled internal procedures.	Public release omits raw datasets, processing scripts, calibration files, and analysis internals.

1. System Role and Delivered Capability

The control and host subsystem provides the supervisory and real-time functions required to operate a coupled dual-motor dynamometer as an engineering characterization platform. The public requirement boundary is defined by observable behavior rather than implementation structure.

- Embedded control remains responsible for time-critical motor regulation.
- The supervisory layer provides bounded commands, status visibility, and acquisition control.
- The system retains evidence needed to distinguish commanded behavior from measured response.
- Public documentation does not disclose algorithms, model topology, source files, signal names, protocol payloads, or tuning values.

2. Control and Supervisory Behavior

The system is treated as a black-box controller with drive, load, enable, feedback, and monitoring functions. Requirements are stated in terms of operator-visible behavior and verification outcomes.

Capability	Public requirement / acceptance basis	Status
Drive/load coordination	One machine establishes shaft motion while the second provides controllable loading or assisting torque.	Demonstrated
Command supervision	Operator commands remain bounded and subject to enable state.	Demonstrated
Feedback visibility	Independent machine feedback is available to support closed-loop operation and review.	Demonstrated
Communications	A supervisory communications path supports command and diagnostic exchange.	Demonstrated

3. Acquisition and Characterization Boundary

The public release states what evidence the platform can produce, but not how acquisition, filtering, binning, calibration, or analysis is implemented. Detailed processing logic, data schemas, scripts, and raw

datasets remain private. Simulink model screenshots are also withheld because readable block topology, signal names, and parameterization can disclose implementation detail even when editable model files are absent.

- Acquisition shall support repeatable comparison of commanded and measured behavior.
- Characterization outputs shall be identified as bench evidence rather than certified metrology.
- Published summaries shall distinguish demonstrated capability from production qualification.

4. Safety and Release Controls

The platform is an engineering test asset. Public documentation therefore defines release boundaries and operator-visible safety expectations without publishing detailed commissioning or wiring procedures.

Control	Acceptance basis	Disposition
Enable state	Torque-producing operation is inhibited when the system is not enabled.	Required
Fault response	A detected control or communications fault produces a bounded response appropriate to a bench prototype.	Required
Public release boundary	No source, model, tuning, protocol, or setup content is distributed.	Verified by package inspection
Qualification claim	No production safety, EMC, environmental, cybersecurity, or endurance certification is claimed.	Explicit limitation

Requirements Verification and Traceability

Verification in this public FRD is intentionally black-box. It establishes whether the externally observable requirement has been demonstrated, inspected, or bounded without exposing source code, models, detailed signal mappings, calibration constants, or setup procedures.

Requirement group	Verification basis	Public disposition
Functional behavior	Bench demonstration and retained internal engineering evidence	Demonstrated where stated
External interfaces	Observed system-level communication and feedback behavior	Demonstrated / bounded
Safety and release controls	Document and operational review	Release-controlled

Requirement group	Verification basis	Public disposition
Performance characterization	Summary-level evidence only; detailed methods retained privately	Capability demonstrated; no certified performance claim

Acceptance and Release Criteria

- All mandatory public functional requirements have a defined verification method and disposition.
- No source code, editable models, setup instructions, tuning data, detailed protocol definitions, or raw datasets are present in the public package.
- Published statements distinguish bench demonstration from calibrated or production-qualified performance.
- Any future public evidence is reviewed against the same black-box release boundary before publication.

Limitations and Document Control

This document is a public black-box FRD. It is not a design specification, source-code release, manufacturing package, laboratory procedure, or deployment guide. Implementation detail sufficient to reproduce, modify, tune, or manufacture the system is intentionally withheld.

Attribution

Herder Elektronische Systemen prepared this public black-box FRD to document engineering capability and verification status. Implementation source, models, protocols, calibration assets, and internal development history are controlled separately and are not part of this release.